

Farm Animals Class Diagrams Program

1. Objective

To design a Java application for farm animals such as Cow, Pig, and Horse and display their place of stay, food habits, and sounds.

2. Algorithm

Start the program.

Create a Cow class with methods whereItStays(), whatItEats(), and sound().

Create a Pig class with the same methods.

Create a Horse class with the same methods.

Create objects of Cow, Pig, and Horse in the main() method.

Call the respective methods for each animal.

Display the details of each animal.

Stop the program.

3. UML Class Diagram

Cow +whereItStays() +whatItEats +sound()
Pig +whereItStays() +whatItEats +sound()
Horse +whereItStays() +whatItEats +sound()
FarmAnimals +main()

4. Source Code

// Class representing a Cow

```
class Cow{
```

```
// Method to display where the cow stays
void whereItStays(){
    System.out.println("Cow stays in a shed.");
}

// Method to display what the cow eats
void whatItEats(){
    System.out.println("Cow eats grass and hay.");
}

// Method to display the sound made by the cow
void sound(){
    System.out.println("Cow makes the sound: Moo");
}
}

// Class representing a Pig
class Pig{
    // Method to display where the pig stays
    void whereItStays(){
        System.out.println("Pig stays in a pigsty.");
    }

    // Method to display what the pig eats
    void whatItEats(){
        System.out.println("Pig eats grains and vegetables.");
    }

    // Method to display the sound made by the pig
    void sound(){
        System.out.println("Pig makes the sound: Oink");
    }
}
```

```

}

// Class representing a Horse
class Horse{

    // Method to display where the horse stays
    void whereItStays(){

        System.out.println("Horse stays in a stable.");

    }

    // Method to display what the horse eats
    void whatItEats(){

        System.out.println("Horse eats grass and hay.");

    }

    // Method to display the sound made by the horse
    void sound(){

        System.out.println("Horse makes the sound: Neigh");

    }

}

// Main class
class FarmAnimals{

    public static void main(String args[]){

        // Creating an object of Cow class
        Cow c = new Cow();

        // Creating an object of Pig class
        Pig p = new Pig();

        // Creating an object of Horse class
        Horse h = new Horse();

        // Displaying Cow details
        System.out.println("----- COW DETAILS -----");
    }

}

```

```
c.whereItStays();
c.whatItEats();
c.sound();
System.out.println();
// Displaying Pig details
System.out.println("----- PIG DETAILS -----");
p.whereItStays();
p.whatItEats();
p.sound();
System.out.println();

// Displaying Horse details
System.out.println("----- HORSE DETAILS -----");
h.whereItStays();
h.whatItEats();
h.sound();
}
}
```

5. Output

----- COW DETAILS -----

Cow stays in a shed.

Cow eats grass and hay.

Cow makes the sound: Moo

----- PIG DETAILS -----

Pig stays in a pigsty.

Pig eats grains and vegetables.

Pig makes the sound: Oink

----- HORSE DETAILS -----

Horse stays in a stable.

Horse eats grass and hay.

Horse makes the sound: Neigh